



Air University, Islamabad

Modules & Functional Requirements

For

EZSell Marketplace Platform

Version 1.1

By

Dawood Haider 221574

Ahmed Abbassi 220965

Muhammad Ahmed 222550

Supervisor

Dr. Madiha Yousaf

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Contents

1. Overview	3
2. Target Users	3
3. Technology Stack	3
4. Modules	4
Module 1: User Management & Authentication	4
Module 2: Product Listings & Marketplace	4
Module 3: AI-Powered Price Prediction	4
Module 4: Smart Recommendations & Search	4
Module 5: Augmented Reality (AR) Try-On	4
Module 6: Admin Analytical Dashboard	4
Module 7: Fraud Prevention	4
Module 8: Messaging & Inbox	5
Module 9: Favorites & User Dashboard	5
Module 10: Support & Notifications	5
5. Functional Requirements	5
Profile Management FR-1	5
Product Listings & Management FR-2	6
Price Prediction FR-3	7
Recommendations FR-4	7
Analytical Dashboard FR-5	8
AR Customization Try-On FR-6	9
Fraud Prevention FR-7	9
Support & Notifications FR-8	10
Messaging & Inbox FR-9	11
Favorites & User Dashboard FR-10	11

1. Overview

EZSell is a full-stack, AI-powered second-hand marketplace platform designed for the Pakistani market. The platform enables users to buy and sell used goods, primarily mobile phones, laptops, and furniture through a secure, intelligent, and visually rich web application. The system is built using a FastAPI backend with a React + Vite frontend, supported by SQLAlchemy for data persistence.

The project scope encompasses all phases of the software development lifecycle, from database design and backend API development to frontend UI/UX, AI/ML model integration, AR visualisation, and an automated fraud prevention pipeline.

2. Target Users

User Type	Description
Buyers	General public purchasing second-hand goods at fair, AI-validated prices
Sellers	Individuals listing used items, assisted by AI price guidance and AR showcasing
Administrators	Platform staff managing listings, users, fraud flags, and support requests via admin console

3. Technology Stack

Layer	Technology
Frontend	React 18, Vite, TypeScript, Tailwind CSS, Shadcn UI, Axios
Backend	FastAPI (Python), SQLAlchemy, Pydantic, Uvicorn
Database	SQLite (development) via SQLAlchemy ORM
Authentication	JWT (python-jose), Google OAuth 2.0
AI / ML	XGBoost, LightGBM, Scikit-learn, SentenceTransformers, Groq API
AR	model-viewer (WebAR), Tripo AI (GLB generation), USDZ conversion
LLM	llama-3.3-70b-versatile via Groq — title validation and price estimation
Market Data	DuckDuckGo DDGS — live OLX Pakistan price scraping
Image Processing	Pillow — perceptual hashing for fraud detection

4. Modules

Module 1: User Management & Authentication

FE-1: User registration with mandatory phone number, email OTP verification, and Google OAuth 2.0 sign-in.

FE-2: JWT-based session management for all secured endpoints.

FE-3: Profile management including avatar upload, bio, location, and contact details.

Module 2: Product Listings & Marketplace

FE-1: Creation, editing, deletion, and approval-gated publishing of product listings.

FE-2: Multi-image upload (minimum 2, maximum 7 per listing; JPG/PNG; max 10MB each).

FE-3: Category-specific attribute fields for Mobiles, Laptops, and Furniture.

FE-4: Listing status controls: active, hidden, sold.

Module 3: AI-Powered Price Prediction

FE-1: Weighted ensemble of four trained ML models (XGBoost, LightGBM, Random Forest, Gradient Boosting) per product category.

FE-2: Real-time market price estimation using Groq LLM (llama-3.3-70b-versatile) with live OLX Pakistan data scraped via DuckDuckGo.

FE-3: NLP feature extraction from listing text; IQR-based outlier filtering on scraped prices.

FE-4: Overall prediction accuracy 85%+ within a +/-20% price margin.

Module 4: Smart Recommendations & Search

FE-1: Activity-based interest profiling per user using SentenceTransformer semantic embeddings.

FE-2: Personalized homepage feed; trending listings for anonymous visitors.

FE-3: Multi-parameter filtering by category, price range, brand, condition, and location.

Module 5: Augmented Reality (AR) Try-On

FE-1: Procedural 3D model generation from listing attributes for furniture items.

FE-2: Tripo AI GLB model generation from listing photos.

FE-3: In-browser WebAR viewer using the model-viewer web component.

FE-4: USDZ export for native iOS AR QuickLook; fixed real-world scale; baked floor shadow decal.

Module 6: Admin Analytical Dashboard

FE-1: Overview analytics: total users, listings, revenue, support tickets.

FE-2: Full listing, user, and approval management with rejection reasons.

FE-3: Support ticket management with status controls and automated user notifications.

Module 7: Fraud Prevention

FE-1: Listing hash deduplication to detect copy-paste duplicate ads.

FE-2: Perceptual image hashing to detect reused or stolen photos.

FE-3: Fraud flag storage and admin review queue.

Module 8: Messaging & Inbox

FE-1: Buyer-to-seller messaging tied to specific listings.

FE-2: Conversation inbox with unread message badge in the navigation bar.

Module 9: Favorites & User Dashboard

FE-1: Per-user saved listings with instant add/remove toggle.

FE-2: Seller personal dashboard showing all ads grouped by status.

Module 10: Support & Notifications

FE-1: User-submitted support requests and bug reports routed to admin panel.

FE-2: Admin status updates trigger automated in-app notifications to the submitting user.

FE-3: Notification bell with unread badge, popover list, and mark-as-read controls.

5. Functional Requirements

Profile Management FR-1

Table 5.1 Functional Requirements FR-1

FR No.	Requirement	Description	Status
FR-M1-1	User Registration	Register with username, email, mandatory phone, and password	Implemented
FR-M1-2	Email Verification	6-digit OTP sent to verify email before account activation	Implemented
FR-M1-3	GoogleOAuth Login	Sign in with Google account via OAuth 2.0	Implemented
FR-M1-4	JWT Authentication	All secured endpoints require a valid JWT Bearer token	Implemented
FR-M1-5	Profile View & Edit	Update full name, bio, phone, location, and avatar	Implemented
FR-M1-6	Password Reset	Request a password reset link via email	Implemented
FR-M1-7	Admin Role	Admin users have elevated privileges to manage listings, users, tickets	Implemented
FR-M1-8	Avatar Upload	Upload a profile picture stored on the backend	Implemented

Product Listings & Management FR-2

Table 5.2 Functional Requirements FR-2

FR No.	Requirement	Description	Status
FR-M2-1	Create Listing	Post ads with title, description, price, category, and images	Implemented
FR-M2-2	Multi-Image Upload	Minimum 2 and up to 7 images per listing (JPG/PNG, max 10MB each)	Implemented
FR-M2-3	AI Title Validation	Title validated via Groq LLM (llama-3.3-70b-versatile) for clarity	Implemented
FR-M2-4	Edit Listing	Owner can update listing details at any time	Implemented
FR-M2-5	Delete Listing	Owner or admin can permanently remove a listing	Implemented
FR-M2-6	Hide / Unhide	Toggle listing visibility without deleting it	Implemented
FR-M2-7	Mark as Sold	Seller marks item as sold; status shown to buyers	Implemented
FR-M2-8	Listing Approval	Submitted listings require admin review before going live	Implemented
FR-M2-9	Category-Specific Fields	Mobiles, Laptops, and Furniture have tailored attribute fields	Implemented
FR-M2-10	Seller Phone on Ad	Buyer can view seller phone number on the listing detail page	Implemented

Price Prediction FR-3

Architecture: XGBoost (35%) + LightGBM (35%) + Random Forest (15%) + Gradient Boosting (15%) | Groq LLM layer with live OLX scraping | Accuracy: 85%+

Table 5.3 Functional Requirements FR-3

FR No.	Requirement	Description	Status
FR-M3-1	AI Price Suggestion	Estimated fair price via blended ML and LLM pipeline	Implemented
FR-M3-2	Category Trained Models	Separate .pkl ensemble models for Mobile, Laptop, and Furniture	Implemented
FR-M3-3	Ensemble ML Models	XGBoost, LightGBM, Random Forest, Gradient Boosting — weighted avg	Implemented
FR-M3-4	Groq LLM Estimation	llama-3.3-70b-versatile fetches live OLX data as a second layer	Implemented
FR-M3-5	Live Market Scraping	DuckDuckGo scrapes OLX Pakistan; IQR outlier filtering applied	Implemented
FR-M3-6	NLP Feature Extraction	Extracts RAM, storage, processor, brand, condition from listing text	Implemented
FR-M3-7	Confidence Score	Prediction returns a confidence % (0-100%) shown to the seller	Implemented
FR-M3-8	Price Range Display	Shows predicted vs. asked price; low-confidence warnings to sellers	Implemented
FR-M3-9	Model Accuracy (>=85%)	85%+ accuracy within +/-20% price margin across all three categories	Achieved

Recommendations FR-4

Table 5.4 Functional Requirements FR-4

FR No.	Requirement	Description	Status
FR-M4-1	Activity Tracking	Logs user views, searches, favorites, and clicks	Implemented

FR-M4-2	Interest Profiling	Aggregated interests computed per user from activity history	Implemented
FR-M4-3	Semantic Embeddings	Listings encoded with SentenceTransformer vectors for similarity search	Implemented
FR-M4-4	Personalized Feed	Homepage shows AI-ranked listings based on user interest profile	Implemented
FR-M4-5	Trending Listings	Anonymous users see trending / popular listings as defaults	Implemented
FR-M4-6	Smart Search & Filter	Filter by category, price range, brand, condition, and location	Implemented

Analytical Dashboard FR-5

Table 5.5 Functional Requirements FR-5

FR No.	Requirement	Description	Status
FR-M5-1	Admin Overview Cards	Dashboard shows total users, listings, revenue, support requests	Implemented
FR-M5-2	Listing Management	Admin can view, approve, reject, and delete any listing	Implemented
FR-M5-3	User Management	Admin can view full user profiles and manage accounts	Implemented
FR-M5-4	Category Analytics	Charts show breakdown of listings by category and approval status	Implemented
FR-M5-5	Engagement Metrics	Tracks views, favorites, and session data per listing	Implemented
FR-M5-6	Support Ticket Dashboard	Admin sees all support/bug tickets with user info and status controls	Implemented
FR-M5-7	Ticket Status Updates	Admin marks tickets as Working or Done from the dashboard	Implemented

AR Customization Try-On FR-6

Table 5.6 Functional Requirements FR-6

FR No.	Requirement	Description	Status
FR-M6-1	Procedural 3D Model	Auto-generates a 3D furniture model from listing attributes	Implemented
FR-M6-2	AI 3D Generation	Tripo AI generates a photorealistic GLB model from listing images	Implemented
FR-M6-3	WebAR Viewer	Users view 3D model in AR via browser using the model-viewer component	Implemented
FR-M6-4	iOS AR QuickLook	USDZ export supports native iOS AR experience	Implemented
FR-M6-5	Fixed Real-World Scale	AR model dimensions locked to real-world measurements in cm	Implemented
FR-M6-6	Floor Grounding	Shadow decal ensures model appears grounded on real surfaces	Implemented
FR-M6-7	Admin GLB Upload	Admin can manually upload a custom GLB file per listing	Implemented
FR-M6-8	AR Coaching Prompts	On-screen overlay guides user to move phone to detect floor	Implemented

Fraud Prevention FR-7

Table 5.7 Functional Requirements FR-7

FR No.	Requirement	Description	Status
FR-M7-1	Duplicate Listing Detection	Hashes title+description+price+owner to detect copy-paste duplicates	Implemented
FR-M7-2	Image Perceptual Hash	Perceptual hash detects reused or stolen listing photos	Implemented
FR-M7-3	Fraud Flag Storage	JSON array of fraud flags stored per listing	Implemented

FR-M7-4	Admin Fraud Review	Flagged listings surfaced to admin for review before approval	Implemented
FR-M7-5	Rejection Reason with	Admin can reject a listing with a documented reason in DB	Implemented

Support & Notifications FR-8

Table 5.8 Functional Requirements FR-8

FR No.	Requirement	Description	Status
FR-M8-1	Contact Support Form	Users submit support or bug reports from their profile page	Implemented
FR-M8-2	Admin Ticket Dashboard	Admin sees all tickets with user profile, type, subject, description	Implemented
FR-M8-3	Ticket Status Control	Admin updates ticket status to Open, Working, or Done	Implemented
FR-M8-4	User Notification	Notification auto-created for user when admin changes ticket status	Implemented
FR-M8-5	Notification Bell Badge	Navbar bell shows combined unread notification and pending item count	Implemented
FR-M8-6	Notification Popover	Dropdown displays up to 5 latest alerts with title, message, and date	Implemented
FR-M8-7	Mark as Read	User can mark individual or all notifications as read	Implemented
FR-M8-8	Notification Link	Clicking a notification redirects user to the relevant section	Implemented

Messaging & Inbox FR-9

Table 5.9 Functional Requirements FR-9

FR No.	Requirement	Description	Status
FR-M9-1	Send Message	Buyers send messages to sellers from the listing detail page	Implemented
FR-M9-2	Inbox / Conversation View	Conversations grouped by listing and contact in an inbox view	Implemented
FR-M9-3	Real-Time Unread Count	Navigation badge shows count of unread messages, updated periodically	Implemented
FR-M9-4	Mark Messages as Read	Opening a conversation marks all messages in it as read	Implemented
FR-M9-5	Seller Contact Info	Seller phone number shown on listing detail page alongside messaging	Implemented

Favorites & User Dashboard FR-10

Table 5.10 Functional Requirements FR-10

FR No.	Requirement	Description	Status
FR-M10-1	Add to Favorites	Heart/save any listing from card or detail page	Implemented
FR-M10-2	Remove from Favorites	Unsave a listing; removed immediately from saved list	Implemented
FR-M10-3	Favorites Page	Dedicated page showing all saved listings with remove option	Implemented
FR-M10-4	Personal Dashboard	Seller dashboard shows active, hidden, sold, pending listings	Implemented
FR-M10-5	Listing Status Counts	Dashboard cards show live counts of active/hidden/sold/pending ads	Implemented
FR-M10-6	Favorites Count in Profile	Profile page shows total saved favorites count as a stat	Implemented